Image Aesthetics Assessment With Attribute-Assisted Multimodal Memory Network

Leida Li, *Member, IEEE*, Tong Zhu, Pengfei Chen, Yuzhe Yang, *Member, IEEE*, Yaqian Li, and Weisi Lin, *Fellow, IEEE*

***Abstract*— Image aesthetics assessment (IAA) has attracted growing interest in recent years but is still challenging due to its highly abstract nature. Nowadays, more and more people tend to comment images shared on the social networks, which can provide rich aesthetics-aware semantic information from different aspects. Therefore, user comments of an image can be exploited as supplementary information for enhancing aesthetic representation learning. Previous researches have demonstrated that aesthetic attributes make significant effect on image aesthetic quality and humans' aesthetic perception. Typically, people are used to give comments on an image from the perspective of aesthetic attributes, based on which the aesthetic quality of images can be inferred. Motivated by this, this paper presents an Attribute-assisted Multimodal Memory Network (AMM-Net) for image aesthetics assessment, which utilizes aesthetic attributes to model the interactions between visual and textual modalities. Specifically, we design two memory networks to capture the attribute-aware information most related to the image and associated comments respectively. Further, with multiple memory hops, attribute semantics shared by the two modalities are refined and cross-modal interactions are enhanced progressively. Finally, more discriminative aesthetic representations can be obtained for IAA. The experimental results and comparisons on two public multimodal IAA datasets demonstrate the superiority of the proposed model over the state-of-the-art methods. The source code is available at https://github.com/zhutong0219/AMM-Net.**

***Index Terms*— Image aesthetics assessment, image quality assessment, multimodal analysis.**

I. Introduction

ALONG with the notable progress of the modern society, human's demands on aesthetics are getting higher and higher. Image aesthetics assessment (IAA) targets at evaluating the aesthetic quality of images by imitating the way of human perception. Recently, IAA has attracted increasing interest from both academia and industry owing to its broad applications in photo album curation [1], smart photography [2], image enhancement [3] and image retrieval [4].

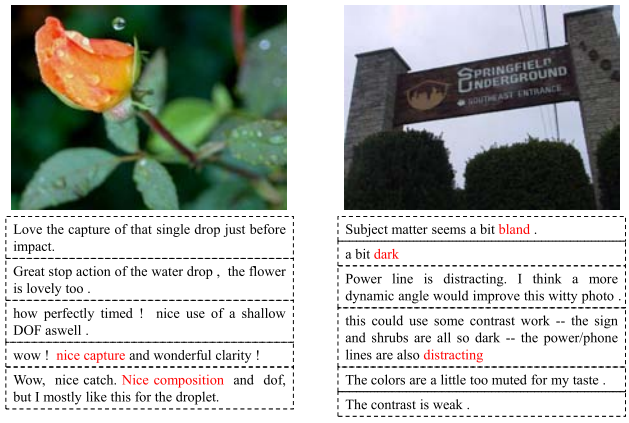

Fig. 1. Example images and the associated comments from the AVA dataset [11]. The left image is with a high aesthetic rating and the right image is with a low one.

Due to its highly abstract nature, IAA remains a challenging research topic. Early IAA models focused on designing hand-crafted features following the photographic rules, *e.g.*, depth of field [5], rule of thirds [6] and color harmony [7]. Recently, with the prevalence of deep learning, Convolutional Neural Networks (CNNs) have been widely utilized in different IAA tasks, *i.e.*, binary classification [8], score regression [9] and distribution prediction [10].

With the boom of mobile internet, more and more users tend to share images and give corresponding comments on social networks. Sometimes, it is challenging to evaluate the aesthetics simply from visual appearances, especially images with mediate aesthetic quality. Fig. 1 shows two example images and associated comments from the AVA dataset [11]. As can be seen, user comments can provide rich semantic information from different perspectives, which facilitates image aesthetic representation learning. For example, the words "nice capture" and "nice composition" suggest that the left image may have a high aesthetic score, while the words "bland", "dark" and "distracting" indicate the potentially low quality of the right image. However, most of the existing IAA methods did not take full advantage of the textual modality. In this paper,

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(a)

(b)

Fig. 2. Example images and the corresponding aesthetic attributes from the AADB dataset [12]. The ranges of aesthetic scores and aesthetic attributes are [1], [5] and [-1, 1] respectively.

we focus on utilizing user comments as auxiliary information by modeling the interactions progressively between visual and textual modalities, aiming for achieving more comprehensive IAA.

The existing works have demonstrated that aesthetic attributes are closely related to image aesthetics and influence human aesthetic perception [9], [12]. Two example images and the associated aesthetic attributes from the AADB dataset [12] are shown in Fig. 2. The ground truth aesthetic score and aesthetic attributes of each image range from 1 to 5 and -1 to 1 respectively. As illustrated in the figure, the image on the top row has high aesthetic quality when aesthetic attributes of the image are prominently positive, and vice versa. In addition, people usually comment an image from the perspective of various aesthetic attributes, *e.g.*, color, composition and content. Therefore, aesthetic attributes can be utilized as a bridge to model the interactive relationships between visual and textual modalities. Intuitively, these attributes determine the aesthetic perception of people, based on which the aesthetic quality of images can be inferred.

Motivated by the above facts, this paper presents an Attribute-assisted Multimodal Memory Network (AMM-Net) for image aesthetics assessment, which progressively captures the most significant aesthetic attribute semantics shared by the visual and textual modalities. Considering that human aesthetic experience is a coarse-to-fine process [13] and aesthetics assessment aims at imitating the way of human perception, we utilize memory network with multiple hops to achieve coarse-to-fine inference of aesthetics based on aesthetic attributes. Specifically, the proposed AMM-Net consists of two memory networks for image and associated comments respectively, modeling the cross-modal interactions with the supervision of the aesthetic attributes. With multiple computational steps (hops), attribute semantics that involve both inter-modal and intra-modal information can be further refined from coarse-to-fine progressively. Finally, more discriminative and accurate aesthetic feature representations are obtained for image aesthetic distribution prediction.

The main contributions can be summarized as follows,

- We propose a new attribute-assisted multimodal memory network for image aesthetics assessment, dubbed AMM-Net. The proposed method not only investigates significant aesthetic attribute semantics related to the visual and textual modalities, but also captures the fine-level interactions between the image and associated comments.
- We design two memory networks for images and corresponding comments respectively to model the cross-modal interactions under the guidance of aesthetic attributes. Multiple memory hops with attention mechanism are also introduced to refine the attribute semantics shared by the two modalities progressively, and this facilitates better aesthetic feature representations.
- We have done extensive experiments on two public multimodal IAA datasets and the results demonstrate that the proposed AMM-Net significantly outperforms the state-of-the-arts for different aesthetics assessment tasks, *i.e.*, binary classification, score regression and distribution prediction.

The remainder of this paper is organized as follows. Section II briefly reviews related works. In Section III, we introduce the architecture of the proposed approach. Experimental results are given and analyzed in Section IV. Finally, we conclude the paper in Section V.

II. Related Work

In this section, we briefly review the existing works on image aesthetics assessment, multimodal learning and memory network, which are closely related to our proposed method.

A. Image Aesthetics Assessment

Early studies on image aesthetics assessment mainly focus on designing handcrafted features. Datta et al. [5] extracted a set of visual features following the photographic rules, *e.g.*, color, rule of thirds and depth of field, which are then employed to train a SVM classifier for image aesthetic quality classification. Wong et al. [14] detected the salient regions of an image, and extracted subject features, subject-background relationships and global features to classify image aesthetic quality. Marchesotti et al. [15] considered generic image descriptors to assess image aesthetics, where Bag-Of-Visual-words (BOV) and Fisher Vector (FV) are used. Tang et al. [16] divided images into seven categories on the basis of visual content. Then, different strategies are designed to extract both regional and global features for aesthetics assessment. While handcrafted features have explicit physical meanings, they are limited in describing the highly abstract nature of image aesthetics.

Recently, with the development of deep learning, convolutional neural networks have been widely used in IAA tasks [17], [18]. Lu et al. [8] leveraged multiple patches randomly cropped from the raw image to categorize image aesthetic quality. To extract object, scene and texture features

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for aesthetics learning, Kao et al. [19] proposed three separate convolutional neural networks for input images. To preserve the original composition information of the input images, Mai et al. [20] adopted an adaptive spatial pooling layer to train the model without any image transformations. Instead of predicting the binary aesthetic quality or mean score, Talebi et al. [21] proposed to predict image aesthetic distribution by using the squared earth movers distance loss. She et al. [10] utilized graph convolutional networks with a hierarchical architecture to capture image layout information for aesthetics assessment. With the assistance of personality data, Li et al. [22] introduced a multi-task learning framework with shared weights for aesthetic distribution prediction. Despite the notable success of deep learning-based IAA methods, most of them only focused on extracting aesthetic information from pure visual modality.

B. Multimodal Learning

Nowadays, with the boom of social media, user-generated contents are increasingly leaning towards the form of multiple modalities [23], [24]. Multimodal learning has been demonstrated more effective compared to single modal-based methods in various tasks [25], [26]. For example, by utilizing mutual influences between visual features and textual words, Zhang et al. [27] proposed a probabilistic semantic model for image annotation and multi-modal image retrieval. He and Peng [28] jointly combined visual and textual information to improve image classification performance. Taking audio context of movies into account, Wu et al. [29] integrated audio and visual features for realistic human action recognition. Zhu et al. [30] captures latent correspondences between image regions and text words for sentiment analysis. Intuitively, data from different modalities can complement and interact with each other [31], which results in advanced performance. However, little attention has been paid on applying multimodal learning for IAA. Therefore, by taking the associated comments as complementary information, we employ aesthetic attributes as a bridge to model the interactions between visual and textual modalities, which facilitates more accurate image aesthetic representation learning.

C. Memory Network

Memory network was first introduced by Weston et al. [32], which performs reasoning with an external memory component that can be easily read and written. Sukhbaatar et al. proposed an extension of the memory network in [33], which allows training in an end-to-end manner with less supervision. Due to its advantage in inference capability, memory network has been successfully adopted for various tasks, *e.g.*, sentiment analysis [34], image-text retrieval [35], one-shot learning [36], object tracking [37] and image generation [38]. With this inspiration, in this paper, we propose a multimodal memory network for IAA, where multiple memory hops are utilized to investigate the most significant aesthetic attribute semantics shared by images and comments progressively.

III. Proposed Method

In this section, we present the details of the proposed Attribute-assisted Multimodal Memory Metwork (AMM-Net) for image aesthetics assessment. Our aim is not only to capture aesthetic attributes most related to visual and textual modalities, but also to investigate the cross-modal aesthetic interactions. The overall architecture of the proposed model is shown in Fig. 3. We first extract aesthetic attribute features of the input image. Then, with the supervision of the aesthetic attributes, two memory networks are exploited for the image and the input comments respectively. We also utilized multiple computational layers (hops) to refine attribute semantics shared by the two modalities and further strengthen the cross-modal interactions. Finally, aesthetic feature representations involving both intra-modal and inter-modal attribute semantics are obtained for aesthetics assessment.

A. Multimodal Data Representation

1) Aesthetic Attribute Feature Embedding: For an input image $\boldsymbol{I}$, we first extract the aesthetic attribute-aware features $\boldsymbol{A}$ by an aesthetic attribute model pre-trained on the AADB dataset [12]. A fully connected layer is then applied to transform $\boldsymbol{A}$ into a set of D-dimensional attribute features $\boldsymbol{F}_a = \{\boldsymbol{a}_i\}_{i=1}^m$, where $\boldsymbol{a}_i \in \mathbb{R}^D$ denotes the feature of the i-th attribute and m is the number of attributes.

2) Visual Memory Representation: Given an input image $\boldsymbol{I}$, the Inception-v3 [39] network pretrained on ImageNet [40] is applied as the backbone to extract a set of visual feature maps $\boldsymbol{m}_j^V$, $j \in [1, L]$, where $L = W \times H$ and W and H denote the width and height of the visual feature map. We refer to each feature map as visual memory pieces and stack them for the external visual memory $\mathbf{M}_V$, which represents all the visual information and is denoted as:

$$\mathbf{M}_V = \{\boldsymbol{m}_1^V, \boldsymbol{m}_2^V, \ldots, \boldsymbol{m}_L^V\}, \boldsymbol{m}_j^V \in \mathbb{R}^D. \tag{1}$$

3) Textual Memory Representation: For the input text $\boldsymbol{T}$ with n words, the pre-trained BERT-Base [41] model is employed to embed each word into a 768-dimensional vector $\boldsymbol{t}_j$, $j \in [1, n]$. GRU is capable of learning long-term dependencies in the sequential information, and bidirectional GRU takes both forward and backward computation. Therefore, to integrate context information into word representations, we use bidirectional GRU [42], [43] to aggregate both forward and backward information in the comments, which is formulated as:

$$\overrightarrow{\boldsymbol{h}_j} = \overrightarrow{GRU}(\boldsymbol{t}_j, \overrightarrow{\boldsymbol{h}_{j-1}}), \; j \in [1, n], \tag{2}$$

$$\overleftarrow{\boldsymbol{h}_j} = \overleftarrow{GRU}(\boldsymbol{t}_j, \overleftarrow{\boldsymbol{h}_{j+1}}), \; j \in [1, n], \tag{3}$$

where $\overrightarrow{\boldsymbol{h}_j}$ and $\overleftarrow{\boldsymbol{h}_j}$ represent the forward hidden state and the backward hidden state respectively. The word representation $\boldsymbol{m}_j^T$ is obtained by calculating the average of bidirectional hidden states as follows:

$$\boldsymbol{m}_j^T = \frac{\overrightarrow{\boldsymbol{h}_j} + \overleftarrow{\boldsymbol{h}_j}}{2}, \; j \in [1, n]. \tag{4}$$

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Fig. 3. The overall framework of the proposed Attribute-assisted Multimodal Memory Network (AMM-Net) for image aesthetics assessment. Aesthetic attribute features are first extracted from the input image. Two memory networks are employed for the image and the input comments respectively. Visual and textual representations with regard to aesthetic attributes of the last hop are then fused for final aesthetics assessment.

We stack the word vectors and regard them as the external textual memory $\mathbf{M}_T$:

$$\mathbf{M}_T = \{\boldsymbol{m}_1^T, \boldsymbol{m}_2^T, \ldots, \boldsymbol{m}_n^T\}, \boldsymbol{m}_j^T \in \mathbb{R}^D. \tag{5}$$

B. Multimodal Memory Network

After obtaining the aesthetic attribute features $\boldsymbol{F}_a$ and the external visual and textual memories ($\mathbf{M}_V$ and $\mathbf{M}_T$), we employ two memory networks to capture significant visual and textual information most related to the given attributes. The proposed AMM-Net consists of multiple computational layers (hops). At the first hop of two memory networks, we introduce the attribute-guided attention mechanism for supervising visual and textual attention with the given aesthetic attributes.

1) Visual Memory Network: The visual memory network is designed to extract informative visual features with regard to aesthetic attribute semantics. In the first memory hop, we take aesthetic attribute features $\boldsymbol{F}_a$ and the external visual memory $\mathbf{M}_V$ as the inputs. For each piece of visual memory $\boldsymbol{m}_j^V$, we compute the semantic dependency with the aesthetic attributes through multilayer perceptron, which is defined as follows:

$$\boldsymbol{p}_{ij}^{(1)} = \tanh(\mathbf{W}_p^{(1)}[\boldsymbol{a}_i, \boldsymbol{m}_j^V] + b_p^{(1)}), \tag{6}$$

where $\mathbf{W}_p^{(1)}$ denotes learnable weight and $b_p^{(1)}$ is a bias term. We employ a softmax function to normalize $\boldsymbol{p}_{ij}^{(1)}$ for attribute-guided visual attention weight, which takes the form:

$$\alpha_{ij}^{(1)} = \text{softmax}(\boldsymbol{p}_{ij}^{(1)}). \tag{7}$$

Finally, to focus on visual contents most related to aesthetic attributes, the output visual vector of the first hop is computed by adaptively assigning an attention weight to each piece of visual memory $\boldsymbol{m}_j^V$ based on the semantic dependency with the aesthetic attributes:

$$\boldsymbol{F}_v^{(1)} = \sum_{j=1}^{L} \alpha_{ij}^{(1)} \boldsymbol{m}_j^V. \tag{8}$$

2) Textual Memory Network: In parallel, the textual memory network is introduced to focus on important words in comments for aesthetics assessment. We input aesthetic attribute features $\boldsymbol{F}_a$ and the external textual memory $\mathbf{M}_T$ in the first hop of the textual memory network. The semantic correlation between words and aesthetic attributes is computed as:

$$\boldsymbol{q}_{ij}^{(1)} = \tanh(\mathbf{W}_q^{(1)}[\boldsymbol{a}_i, \boldsymbol{m}_j^T] + b_q^{(1)}), \tag{9}$$

where $\mathbf{W}_q^{(1)}$ and $b_q^{(1)}$ are learnable parameters during training. Then attribute-guided textual attention weight is normalized by a softmax function as:

$$\beta_{ij}^{(1)} = \text{softmax}(\boldsymbol{q}_{ij}^{(1)}). \tag{10}$$

Finally, the textual vector $\boldsymbol{F}_t^{(1)}$ is calculated as a weighted sum of each piece of textual memory $\boldsymbol{m}_j^T$ to pay attention to

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informative words related to aesthetic attributes:

$$\boldsymbol{F}_t^{(1)} = \sum_{j=1}^{n} \beta_{ij}^{(1)} \boldsymbol{m}_j^T. \tag{11}$$

C. Cross-Modal Multiple Memory Attention

As aforementioned, visual and textual modalities can influence and complement each other in aesthetics assessment. To fully explore the cross-modal interactions between the image and corresponding comments, we propose a cross-modal multiple memory attention mechanism, which considers both inter-modal attention and intra-modal attention.

1) Visual Memory Attention: With the obtained attribute-guided visual feature vector $\boldsymbol{F}_v^{(1)}$ and attribute-guided textual feature vector $\boldsymbol{F}_t^{(1)}$, we incorporate the visual feature vector $\boldsymbol{F}_v^{(k-1)}$ with the visual memory $\mathbf{M}_V$ to learn the self-influence within the visual modality for the k-th hop of the visual memory network. Similar to the above attribute-guided attention mechanism, we formulate the operations as follows:

$$\boldsymbol{p}_{ij}^{(k)} = \tanh(\mathbf{W}_p^{(k)}[\boldsymbol{F}_v^{(k-1)}, \boldsymbol{m}_j^V] + b_p^{(k)}), \tag{12}$$

$$\alpha_{ij}^{(k)} = \text{softmax}(\boldsymbol{p}_{ij}^{(k)}), \tag{13}$$

$$\boldsymbol{F}_{v2v}^{(k)} = \sum_{j=1}^{L} \alpha_{ij}^{(k)} \boldsymbol{m}_j^V, \tag{14}$$

where $k \in [2, h]$ and h denotes the number of memory hop.

Then, we employ the textual feature vector $\boldsymbol{F}_t^{(k-1)}$ with the visual memory $\mathbf{M}_V$ to capture cross-modal influences that comments bring to the image:

$$\boldsymbol{F}_{t2v}^{(k)} = AAM[\boldsymbol{F}_t^{(k-1)}, \boldsymbol{m}_j^V], \tag{15}$$

where AAM refers to the above operations, *i.e.*, Eq. (12)-(14).

Finally, we take the average of $\boldsymbol{F}_{v2v}^{(k)}$ and $\boldsymbol{F}_{t2v}^{(k)}$ for the attribute-guided visual representation features of the k-th memory hop:

$$\boldsymbol{F}_v^{(k)} = \frac{\boldsymbol{F}_{v2v}^{(k)} + \boldsymbol{F}_{t2v}^{(k)}}{2}. \tag{16}$$

2) Textual Memory Attention: Simultaneously, in the k-th hop of the textual memory network, we also consider both self-influences in the textual modality and inter-modal influences that the image brings to text. The k-th attribute-guided textual feature vector is computed as follows:

$$\boldsymbol{F}_{t2t}^{(k)} = AAM[\boldsymbol{F}_t^{(k-1)}, \boldsymbol{m}_j^T], \tag{17}$$

$$\boldsymbol{F}_{v2t}^{(k)} = AAM[\boldsymbol{F}_v^{(k-1)}, \boldsymbol{m}_j^T], \tag{18}$$

$$\boldsymbol{F}_t^{(k)} = \frac{\boldsymbol{F}_{t2t}^{(k)} + \boldsymbol{F}_{v2t}^{(k)}}{2}. \tag{19}$$

3) Multiple Memory Hops: Considering that one memory layer is limited in learning the sophisticated nature of aesthetics, we stack the proposed attention mechanism with multiple memory hops to learn aesthetic representations with multi-level abstraction. The output of visual and textual attention layer is updated as the input of the next memory hop respectively through a linear layer. Each hop explores significant image contents or sentence words and progressively refines the representation of the previous hop to a higher level. We denote visual and textual representations with regard to aesthetic attributes of the last hop as $\boldsymbol{F}_v^{(h)}$ and $\boldsymbol{F}_t^{(h)}$, which are combined together for the final aesthetics assessment task:

$$\boldsymbol{F} = \boldsymbol{F}_v^{(h)} \oplus \boldsymbol{F}_t^{(h)}, \tag{20}$$

where $\oplus$ represents the concatenation operation.

Based on the above operations, the alignment between visual memory network and the textual memory network can be realized. First, we extract aesthetic attribute features and utilize the attributes to guide the first hop of the visual and textual memory networks respectively, which captures significant visual and textual information most related to the given aesthetic attributes, and thus achieves the first-level alignment between visual and textual memory networks. Then, we propose the multiple computational hops based on cross-modal attention mechanism for two memory networks, which further incorporates the cross-modal information together and captures the influences that one modality brings to the other. This can be seen as the second-level alignment.

D. Aesthetic Distribution Prediction

After obtaining the feature $\boldsymbol{F}$ that captures both aesthetic attribute semantics shared by two modalities and cross-modal interactions, we apply a fully connected (FC) layer and a softmax activation function to predict the aesthetic distribution, which is defined as:

$$\hat{\boldsymbol{y}} = \text{softmax}(\text{FC}(\boldsymbol{F})). \tag{21}$$

The ground truth distribution is denoted as $\boldsymbol{y} = \{y^{(1)}, y^{(2)}, \ldots, y^{(N)}\}$, where $\sum_{k=1}^{N} y^{(k)} = 1$ and N denotes the number of score bins. Following the previous works [10], [21], our model is optimized by minimizing the EMD (earth mover's distance) loss, which is defined as:

$$\mathcal{L}_{EMD} = \left(\frac{1}{N} \sum_{k=1}^{N} |CDF_y(k) - CDF_{\hat{y}}(k)|^r \right)^{\frac{1}{r}}, \tag{22}$$

where $CDF_y(k) = \sum_{k=1}^{N} y^{(k)}$ represents the cumulative distribution function.

During the testing phase, our proposed model can directly predict the aesthetic distribution, and the aesthetic quality score and binary classification results can be calculated consequently.

IV. Experiments

In this section, we perform extensive experiments and comparisons to demonstrate the effectiveness of the proposed AMM-Net model on two public multimodal aesthetic datasets, *i.e.*, AVA [11] and Photo.net [5].

A. Datasets

AVA dataset [11] consists of more than 250,000 images collected from DPChallenge. Each image is rated by approximately 210 users with the aesthetic scores ranging from 1 to

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10 and 10 is the highest aesthetic score. The corresponding comments for each image is provided by AVA-COMMENTS [44]. Following the previous methods [10], [22], we use the same splits with 235,510 images for training and 19,998 images for testing.

Photo.net dataset [5] contains 17,201 images with aesthetic distribution labels, and the score ratings range from 1 to 7. Zhang et al. [45] crawled user comments for images in Photo.net. We use the same data partitions as [46], where 14,613 images are used for training, 974 images are used for validation, and the rest 1,170 images are used for testing.

AADB dataset [12] consists of 10,000 images collected from Flickr. Each image is rated by five people on average with an aesthetic score and eleven aesthetic attributes, including interesting content, object emphasis, good lighting, color harmony, vivid color, depth of field, motion blur, rule of thirds, balancing element, repetition and symmetry. In this work, we utilize the model pre-trained on the AADB dataset to extract aesthetic attributes of images in AVA and Photo.net.

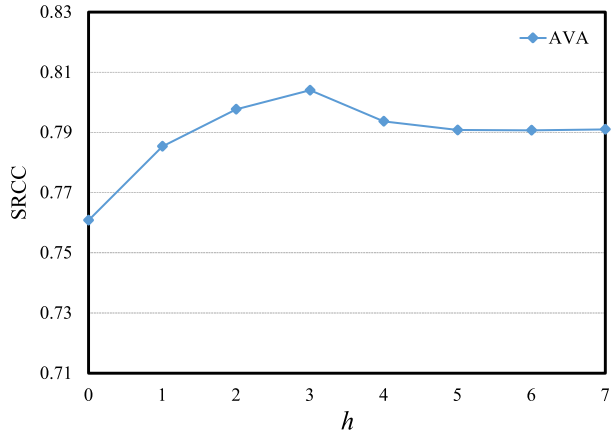
B. Experimental Settings

1) Implementation Details: The Inception-v3 pretrained on ImageNet is used as the backbone. For comparison, we also utilize the Swin Transformer [52] as the backbone to process the input images. We optimize the model by minimizing the EMD loss with $r = 2$ applying stochastic gradient descent (SGD). The initial learning rate is set to 0.01 and drops by a factor of 0.1 every 10 epochs. We apply the weight decay of $1e-5$ with a momentum of 0.9. The batch size is set to 32. We extract eleven aesthetic attributes of each image so $m = 11$. The number of score bins N is equal to 10 for AVA dataset and 7 for Photo.net dataset respectively. Our framework is implemented by PyTorch [53]. All experiments are performed on a NVIDIA GTX 1080Ti GPU.

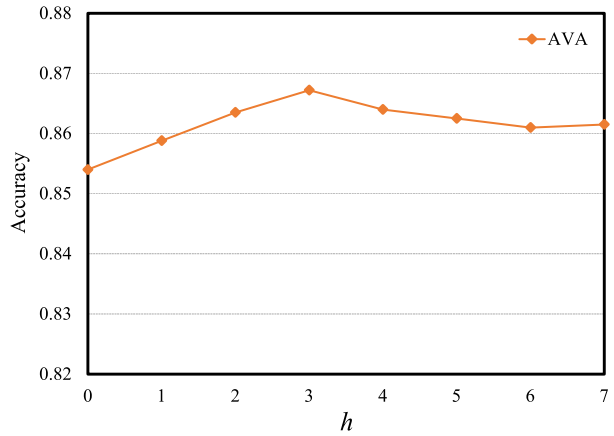
2) Evaluation Criteria: We employ five commonly used metrics to evaluate the performance of the proposed method for three IAA tasks. Accuracy (ACC) is employed for binary aesthetic classification. For aesthetic score regression, Spearman Rank-order Correlation Coefficient (SRCC), Pearson Linear Correlation Coefficient (PLCC) and Mean Squared Error (MSE) are adopted. The Earth Mover's Distance (EMD) is leveraged to evaluate the performance of aesthetic distribution prediction.

C. Comparison With State-of-the-Art Methods

We compare the proposed AMM-Net with the state-of-the-art IAA models on AVA and Photo.net datasets, and the experimental results are summarized in Table I, Table II and Table III respectively. The best performance of each metric is marked in boldface. "-" denotes the unreported metrics in the original papers. For AVA dataset, representative single-modal approaches and up-to-date multimodal methods are compared. From Table I, it can be observed that the proposed AMM-Net is superior to all image modality-based approaches, which is mainly due to the introduction of textual information. Moreover, as can be seen in Table II, among the multimodal-based methods, Multimodal DBM [44] and MultiGAP + RNN [51] simply fuse visual and textual features for aesthetic classification and deliver inferior results. With fully exploring cross-modal interactions, our proposed AMM-Net performs better than Niu et al. [2].



(a) SRCC performance of AMM-Net with different h values



(b) Accuracy performance of AMM-Net with different h values

Fig. 4. Hyperparameter Analysis of different numbers of memory hops for proposed AMM-Net.

Taking into consideration that the IAA approaches experimented on Photo.net dataset are relatively few, we only compare three representative models, including two single-modal methods and one multimodal-based method (MRACNN [45]). From Table III, we can see that the proposed AMM-Net consistently outperforms the state-of-the-art methods on Photo.net in terms of all metrics for binary classification, score regression and distribution prediction.

In general, the experimental results demonstrate that our model can not only model significant aesthetic attribute semantics shared by two modalities, but also explore complicated cross-modal interactions with stacked memory networks, which facilitates more accurate and more comprehensive aesthetic representations for IAA tasks.

D. Ablation Studies

To further demonstrate the effectiveness of each component of our proposed model, we conduct several ablation experiments on AVA dataset in this section. Table IV summarizes the experimental results. To be specific, "Image" refers to the base-

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TABLE I

COMPARISON WITH THE STATE-OF-THE-ART IMAGE-BASED METHODS ON AVA DATASET. "-" DENOTES THE UNREPORTED METRICS IN THE ORIGINAL PAPERS

Methods	Backbone	Acc(%)↑	SRCC↑	PLCC↑	MSE↓	EMD↓
DMA-Net [8]	AlexNet	75.41	-	-	-	-
MNA-CNN [20]	VGG16	77.10	-	-	-	-
MTCNN [47]	VGG16	78.46	-	-	-	-
Kong *et al.* [12]	AlexNet	77.33	0.558	-	-	-
Zeng *et al.* [48]	ResNet-101	80.81	0.719	0.720	0.275	-
APM [49]	ResNet-101	80.30	0.709	-	0.279	-
NIMA [21]	Inception-v2	81.51	0.612	0.636	-	0.050
GPF-CNN [46]	Inception-v3	81.81	0.690	0.704	0.275	0.045
AFDC [50]	ResNet-50	83.24	0.649	0.671	0.271	0.045
PA_IAA [22]	Inception-v3	83.70	0.677	-	-	0.047
HLA-GCN [10]	ResNet-101	84.60	0.665	0.687	0.255	0.043
AMM-Net	Inception-v3	**86.72**	**0.804**	**0.821**	**0.166**	**0.036**
AMM-Net	Swin Transformer	**87.58**	**0.816**	**0.830**	**0.160**	**0.035**

TABLE II

COMPARISON WITH THE STATE-OF-THE-ART MULTIMODAL-BASED METHODS ON AVA DATASET. "-" DENOTES THE UNREPORTED METRICS IN THE ORIGINAL PAPERS

Methods	Backbone	Acc(%)↑	SRCC↑	PLCC↑	MSE↓	EMD↓
Multimodal DBM [44]	VGG16	78.88	-	-	-	-
SingleGAP + RNN [51]	Inception-v1	80.54	-	-	-	-
MultiGAP + RNN [51]	Inception-v1	82.27	-	-	-	-
Niu *et al.* [2]	ResNet-50	81.86	0.734	0.740	0.242	-
AMM-Net	Inception-v3	**86.72**	**0.804**	**0.821**	**0.166**	**0.036**
AMM-Net	Swin Transformer	**87.58**	**0.816**	**0.830**	**0.160**	**0.035**

TABLE III

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON PHOTO.NET DATASET. "-" DENOTES THE UNREPORTED METRICS IN THE ORIGINAL PAPERS

	Methods	Backbone	Acc(%)↑	SRCC↑	PLCC↑	MSE↓	EMD↓
Image-based	MTCNN [47]	VGG16	65.20	-	-	-	-
	GPF-CNN [46]	VGG16	75.60	0.522	0.546	0.272	0.070
	GPF-CNN [46]	Inception-v3	77.19	0.502	0.525	0.252	0.071
Multimodal-based	MRACNN [45]	VGG16	78.91	0.571	0.590	0.211	0.062
	AMM-Net	Inception-v3	**81.90**	**0.674**	**0.678**	**0.197**	**0.061**
	AMM-Net	Swin Transformer	**83.85**	**0.693**	**0.705**	**0.171**	**0.060**

TABLE IV

ABLATION EXPERIMENTS OF THE PROPOSED AMM-NET ON AVA DATASET

Image	Text	Attribute	Memory-V	Memory-T	Sum	Product	Concat	Acc(%)↑	SRCC↑	PLCC↑	MSE↓	EMD↓
✓								83.20	0.621	0.643	0.290	0.047
✓	✓							84.59	0.748	0.765	0.201	0.040
✓	✓	✓						85.47	0.761	0.778	0.196	0.039
✓	✓	✓	✓					86.20	0.782	0.800	0.180	0.038
✓	✓	✓	✓	✓	✓			86.34	0.793	0.812	0.176	0.037
✓	✓	✓	✓	✓		✓		85.23	0.769	0.786	0.182	0.038
✓	✓	✓	✓	✓			✓	**86.72**	**0.804**	**0.821**	**0.166**	**0.036**

line model, which directly applies global visual features to predict aesthetic distribution. On the basis of the baseline, "Text" that refers to the corresponding comments is incorporated for aesthetics assessment. Then, "Attribute" denotes the addition of extracted aesthetic attribute features. "Memory-V" and "Memory-T" refer to visual memory network and textual

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(a) Example images from the AVA dataset

(b) Example images from the Photo.net dataset

Fig. 5. Example images from the AVA and Photo.net dataset tested by our proposed AMM-Net. The ground truth (GT) and predicted (Pre) aesthetic distributions are shown below each image. The average scores are also shown.

memory network respectively. Furthermore, "Sum", "Product" and "Concat" denote different manners, *i.e.*, feature summation, Hadamard product [54] and feature concatenation, to combine the final visual and textual representations respectively.

From Table IV, we can observe that: (1) the proposed AMM-Net comprising of all components achieves the best performance on AVA dataset. Removing any component would lead to suboptimal results; (2) the model with comments is better than the baseline model, demonstrating the effectiveness of utilizing complementary text information; (3) compared with the model with image and text, the superior results of AMM-Net combined with aesthetic attributes indicate that attribute semantics are beneficial for aesthetic representations; (4) based on the above model, the visual memory network is introduced and achieves better performances, which verifies the effectiveness of aesthetic attribute semantics shared by two modalities; (5) from the comparison results of AMM-Net and AMM-Net without the textual memory network, it is obvious that the reinforcement of cross-modal interactions significantly benefits the final aesthetic representations with more information; and (6) feature concatenation achieves the best performance among the three fusion manners, so we choose concatenation in our model to combine multimodal features for the final aesthetic assessment. From the above analysis, we can conclude that each component is indispensable and corporately contributes to the final result.

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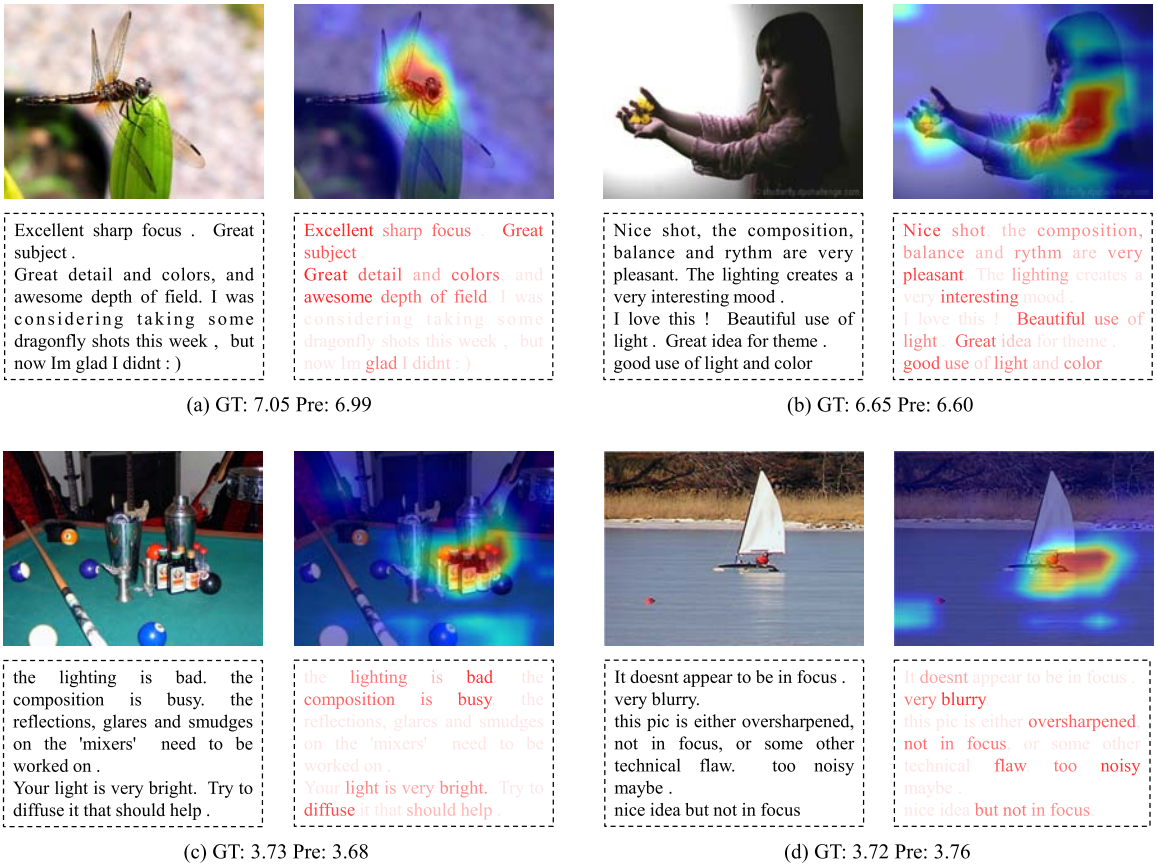

Fig. 6. The visualization of visual and textual attention learned by our proposed AMM-Net. The top row presents images with high aesthetic quality and the corresponding user comments, and the bottom row gives low-quality ones. The bigger the weight of the word, the heavier its color.

E. Hyperparameter Analysis

Considering that the memory hop number h is a key hyperparameter, we conduct experiments on AVA dataset when h is set to various values. Fig. 4 illustrates the experimental results of the proposed AMM-Net under different hop numbers. Here, $h = 0$ denotes the model without employing visual and textual memory networks, and its worst performance further demonstrates that more accurate aesthetic representations can be learned with our proposed method. In addition, $h = 1$ indicates no stacked memory operation and only uses one memory hop, which leads to inferior results. Therefore, it is insufficient to capture attribute semantics and represent multimodal data. We can observe that the SRCC shows a rising trend on the whole when h varies from 0 to 3. However, after $h = 3$, the SRCC begins to drop, which may be due to the redundancy information with the increasing number of computational hops. The accuracy performance is similar to the SRCC. In implementation, we set the number of memory hops $h = 3$ for all experiments in this paper.

F. Visualization

To have an intuitive understanding of the proposed AMM-Net for unified aesthetics assessment, Fig. 5 shows the visualized results of some example images from the AVA and Photo.net dataset respectively. As can be observed from the figure, the predicted aesthetic distributions are highly consistent with the ground truth on two datasets, regardless of whether the aesthetic quality of the image is high, medium or low. The proposed method can learn effective aesthetic representations for aesthetic distribution prediction.

To better understand the rationality of our method, we also visualize aesthetics-specific activation by extracting class activation maps [55] from fine-tuned AMM-Net in Fig. 6. Each row shows the input images and the corresponding user comments as well as the their visualized attention results. The top row presents image-text pairs with high aesthetic quality, and the bottom row gives low-quality ones. The bigger the weight of the word, the heavier its color. From Fig. 6, we can observe that our model is able to pay attention to the significant regions in images, and the proposed aesthetic attributes-assisted mechanism helps focus on important words that benefit for aesthetics assessment. For example, the textual comments in Fig. 6(a) focus on words like “Excellent focus” and “Great detail”, which indicate potentially high aesthetic quality of the image. On the contrary, for the bottom row, words such as “bad lighting”, “busy composition” and “blurry” are mostly paid attention to, suggesting that Fig. 6(c) and Fig. 6(d) tend to have low aesthetic scores. Therefore, the above observation further verifies that our proposed AMM-Net can obtain more accurate aesthetic representations which are beneficial to aesthetics assessment.

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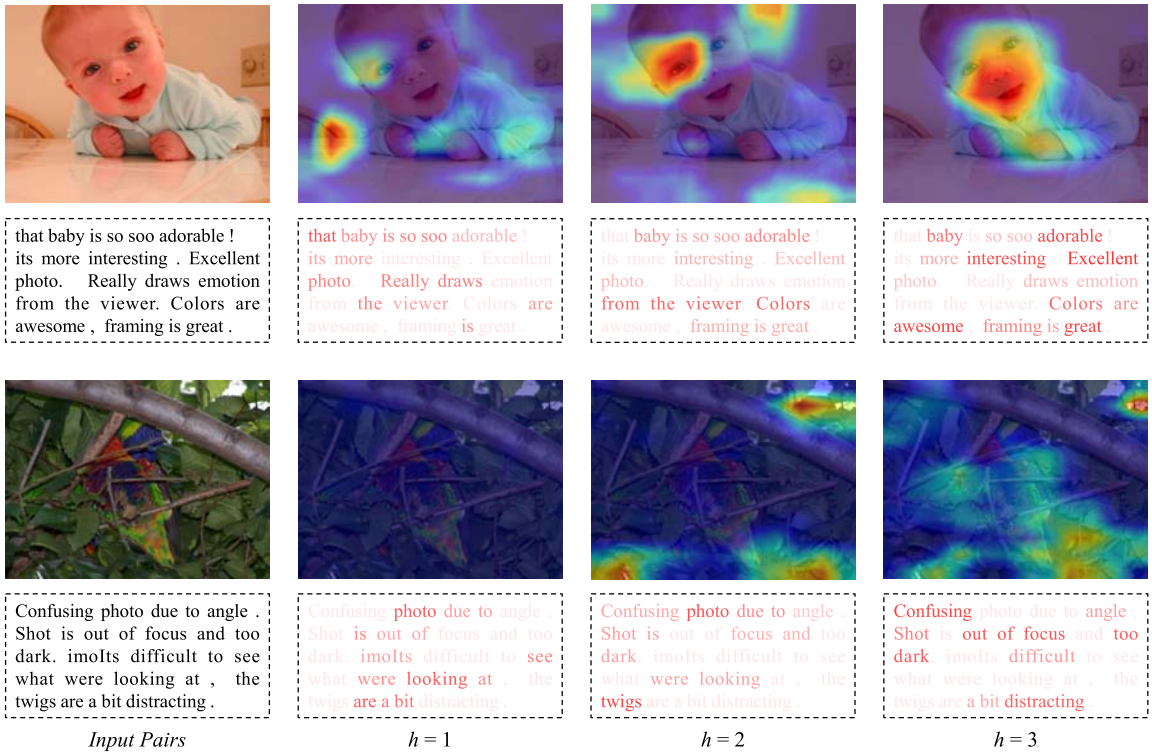

Fig. 7. The visualization of visual and textual attention based on different number of memory hops. The first column presents the original images and comments, and h denotes the number of memory hops.

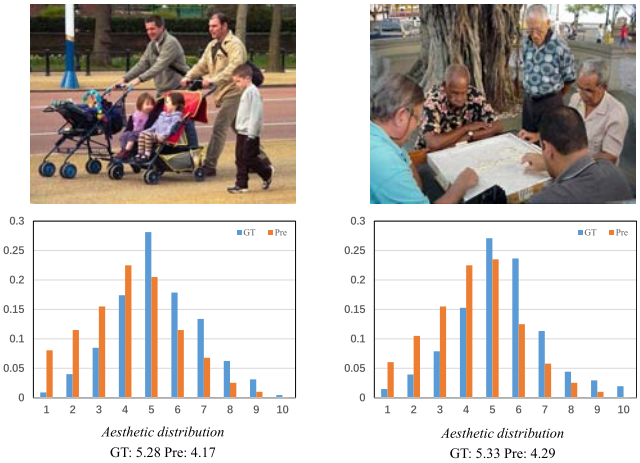

Fig. 8. Two failure examples of the proposed AMM-Net on the AVA dataset. The ground truth (GT) and predicted (Pre) aesthetic distributions as well as scores are shown below each image. Images with scores higher than 5 are considered as high-level aesthetics, while the predicted scores are smaller than 5 and belong to low-level aesthetics.

Fig. 7 shows example activation maps obtained by AMM-Net based on different number of memory hops. The top example is with a high aesthetic rating and the bottom is with a low one. As can be seen from Fig. 7, with the increase of h, the learned attention is able to be refined from coarse-to-fine progressively. The proposed model is able to focus on the most important regions in the image and the most significant words of the comments with $h = 3$, which in turn demonstrates the rationality of the proposed AMM-Net.

G. Limitation

While the proposed AMM-Net has demonstrated its advantage in multimodal IAA, it also has some weakness. Specifically, since AMM-Net is based on aesthetic attributes, it is more suitable for images with significant aesthetic attributes, i.e., dominant attributes can be easily found in the image. For images without dominant aesthetic attributes, the proposed model may produce inaccurate predictions. We show two example images of this kind and the predicted aesthetic distributions in Fig. 8. Images with scores higher than 5 are considered as high-level aesthetics, while the predicted scores are smaller than 5 and belong to low-level aesthetics. It can be observed from the figure that there are no prominent aesthetic attributes in the two images. In this case, the proposed model makes predictions only based on the respective visual and textual features (cross-modal interaction is not effective in this case), and the aesthetic predictions are not very accurate.

V. Conclusion and Future Work

In this paper, we have proposed an attribute-assisted multimodal memory network (AMM-Net) for unified image aesthetics assessment. The proposed visual and textual memory networks capture significant aesthetic attribute semantics shared by the two modalities. In addition, with multiple memory hops, attribute semantics are further refined and cross-modal interactions are strengthened progressively, which benefit more discriminative aesthetic feature representation learning. Extensive experimental results and comparisons demonstrate that the proposed AMM-Net outperforms the state-of-the-art methods.

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The key to multimodal image aesthetics assessment is to capture the cross-modal interaction between the visual and textual modalities. Contrastive Language-Image Pre-training (CLIP) [56] has demonstrated its advantage in exploiting the cross-modality interaction, and has been widely adopted for downstream tasks. As future work, it would be interesting to investigate the use of CLIP in multimodal IAA.

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